

Assignment: Advanced IP Subnetting Scenarios

Instructions: Read each scenario carefully. For each situation, calculate the new subnet mask (in both CIDR and decimal formats), the number of usable hosts per subnet, and the specific network ranges requested. Show your work.

Part 1: Class C Subnetting Scenarios

Scenario 1: The Small Law Firm

The Situation: A local law firm has been assigned the Class C network block **192.168.15.0/24**. The IT director needs to segment the network to separate traffic for security purposes. They need exactly 4 subnets: one for Partners, one for Associates, one for Paralegals, and one for Guest Wi-Fi. **Student Tasks:**

- A) What is the new subnet mask required to create exactly 4 subnets? (Provide CIDR and Decimal).
- B) How many usable host IP addresses are available in each subnet?
- C) Provide the Network ID, Usable IP Range, and Broadcast Address for the **2nd subnet** (Associates).

A) The New Subnet Mask

By borrowing 2 bits from the original /24 mask, your new prefix becomes /26.

CIDR: /26

Dotted Decimal: 255.255.255.192

B) Usable Host IP Addresses

To find the number of hosts, we look at the remaining 6 bits in the last octet ($32 - 26 = 6$). The formula for usable hosts is $2^n - 2$, where n is the number of host bits.

$2^6 = 64$ total addresses per subnet.

$64 - 2 = 62$ usable host IP addresses per subnet.

(We subtract 2 because the first address is the Network ID and the last is the Broadcast address.)

C) Details for the 2nd Subnet (Associates)

The subnets go up in increments of 64. Here is the breakdown for the Associates segment:

Network ID	192.168.15.64
Usable IP Range	192.168.15.65 to 192.168.15.126
Broadcast Address	192.168.15.127

Scenario 2: The Medical Clinic

The Situation: A medical clinic is upgrading its network using the block **200.50.25.0/24**. To comply with healthcare data regulations, they must isolate different departments. They need a minimum of 6 subnets (Reception, Doctors, Nurses, X-Ray, Admin, Pharmacy). **Student Tasks:**

- A) How many network bits must you borrow to accommodate at least 6 subnets, and how many total subnets will this create?
- B) What is the new subnet mask? (Provide CIDR and Decimal).
- C) Provide the Network ID and Broadcast Address for the **1st subnet** (Reception).

A) Borrowed Bits and Total Subnets

To get at least 6 subnets, we need to borrow 3 bits from the host portion.

Formula: $2^3 = 8$

Total Subnets Created: Borrowing 3 bits actually creates 8 subnets. This gives the clinic two extra subnets for future expansion (like a "Lab" or "Billing" department).

B) New Subnet Mask

Since we started with a /24 and added 3 borrowed bits, our new prefix is 27.

CIDR: /27

Dotted Decimal: 255.255.255.224

Note: The value 224 comes from adding the binary place values of the borrowed bits ($128 + 64 + 32 = 224$).

C) 1st Subnet Details (Reception)

The first subnet always starts at the beginning of the assigned block. With a /27 mask, the network increments (the "magic number") in blocks of 32.

Network ID 200.50.25.0

First Usable IP 200.50.25.1

Last Usable IP 200.50.25.30

Broadcast Address 200.50.25.31

Scenario 3: The High School Computer Labs

The Situation: A high school is setting up 4 new computer labs. They have the IP block **192.168.100.0/24**. The principal mandates that each of the 4 labs must be on its own subnet and must be able to support at least 50 computers. **Student Tasks:**

- A) What subnet mask will allow for 4 subnets that each support at least 50 usable hosts? (Provide CIDR and Decimal).
- B) What is the maximum number of usable hosts this mask actually provides per subnet?
- C) Provide the Usable IP Range for the **4th subnet**.

A) The Required Subnet Mask

Since we are starting with a /24 and borrowing 2 bits, our new prefix is /26.

CIDR: /26

Dotted Decimal: 255.255.255.192

This is the only mask that satisfies both requirements. If you borrowed 3 bits (/27), you would have 8 subnets, but only 30 usable hosts per lab—which would fail the principal's mandate of 50 computers.

B) Maximum Usable Hosts

With a /26 mask, we have 6 bits remaining for hosts ($32 - 26 = 6$).

Total Addresses: $2^6 = 64$

Usable Hosts: $64 - 2 = 62$ usable host IP addresses

This provides a nice "buffer" of 12 extra IP addresses per lab for things like printers, teacher stations, or wireless access points.

C) Usable IP Range for the 4th Subnet

To find the 4th subnet, we count in increments of 64 (the "magic number").

Subnet 1: .0 to .63

Subnet 2: .64 to .127

Subnet 3: .128 to .191

Subnet 4: .192 to .255

For the 4th Subnet:

Network ID: 192.168.100.192

Usable IP Range: 192.168.100.193 to 192.168.100.254

Broadcast Address: 192.168.100.255

Scenario 4: The Coffee Shop Chain

The Situation: A regional coffee shop uses the network **198.100.50.0/24** for its corporate office. The network engineer wants to divide the network into the maximum number of subnets possible, provided that every subnet can support at least 14 usable hosts (for point-of-sale systems and back-office PCs). **Student Tasks:**

- A) What is the most efficient subnet mask to achieve this? (Provide CIDR and Decimal).
- B) How many total subnets does this mask create?
- C) Provide the Network ID for the **3rd subnet**.

A) The Most Efficient Subnet Mask

To support 14 usable hosts, we need a host portion large enough to satisfy the formula $2^n - 2 \geq 14$.

If we use 3 bits ($2^3 - 2 = 6$), it's too small.

If we use 4 bits ($2^4 - 2 = 14$), it fits exactly.

Since we use 4 bits for hosts, the remaining 28 bits ($32 - 4 = 28$) are used for the network.

CIDR: /28

Dotted Decimal: 255.255.255.240

B) Total Subnets Created

To find the number of subnets, we look at how many bits were borrowed from the original Class C (/24) block. We moved from /24 to /28, meaning we borrowed 4 bits.

Formula: $2^4 = 16$

Total Subnets: This mask creates 16 total subnets.

C) Network ID for the 3rd Subnet

In a /28 network, the "magic number" (block size) is 16. To find the 3rd subnet, we simply count by 16s starting from zero:

Subnet 1: 198.100.50.0

Subnet 2: 198.100.50.16

Subnet 3: 198.100.50.32

Network ID for the 3rd Subnet: 198.100.50.32

Summary of the 3rd Subnet (Internal View)

Network ID: 198.100.50.32

Usable IP Range: 198.100.50.33 to 198.100.50.46

Broadcast Address: 198.100.50.47

Scenario 5: The E-Commerce Startup

The Situation: A fast-growing tech startup is using the block **203.0.113.0/24**. They have adopted a "micro-team" structure and need to divide their network into 12 distinct subnets for different development squads. **Student Tasks:**

- A) How many bits must be borrowed to create at least 12 subnets?
- B) How many usable IP addresses will be left for each development squad?
- C) Provide the Network ID, Usable IP Range, and Broadcast Address for the **last (highest)** available subnet.

A) Borrowed Bits and Total Subnets

To accommodate at least 12 subnets, you must borrow 4 bits from the host portion of the /24 block.

Total Subnets Created: $2^4 = 16$.

Result: This fulfills the requirement for 12 squads and leaves 4 "spare" subnets for future growth or specialized services (like a dedicated DevOps or Security subnet).

B) Usable IP Addresses per Subnet

By borrowing 4 bits, we are left with 4 bits for hosts ($32 - 28 = 4$). To find the usable IPs, we use the $2^n - 2$ formula:

Total Addresses: $2^4 = 16$.

Usable Host IPs: $16 - 2 = 14$ usable IP addresses per squad.

This is perfect for small "micro-teams" consisting of a few workstations, a local testing server, and a printer.

C) Details for the Last (16th) Subnet

With a /28 mask, the "magic number" (the increment) is 16. The subnets start at .0 and go up to the final block starting at .240.

For the 16th Subnet:

Network ID: 203.0.113.240

Usable IP Range: 203.0.113.241 to 203.0.113.254

Broadcast Address: 203.0.113.255

Part 2: Class B Subnetting Scenarios

Scenario 6: The Regional Hospital

The Situation: A large hospital operates on the Class B network **172.16.0.0/16**. The IT department needs to create 50 separate subnets to isolate different floors, surgical wards, and life-support machine networks. **Student Tasks:**

- A) How many bits must be borrowed to create at least 50 subnets, and what is the new subnet mask? (Provide CIDR and Decimal).
- B) How many usable hosts will each of these new subnets support?
- C) Provide the Network ID and Broadcast Address for the **2nd subnet**.

A) Borrowed Bits and Subnet Mask

To accommodate 50 subnets, we find the smallest power of 2 that is greater than or equal to 50. Since $2^5 = 32$ (too small) and $2^6 = 64$ (perfect), we must borrow 6 bits.

Bits Borrowed: 6 bits

New CIDR: /22 ($16 + 6 = 22$)

Dotted Decimal Mask: 255.255.252.0

Note: The 252 in the third octet comes from the sum of the borrowed bit values ($128 + 64 + 32 + 16 + 8 + 4 = 252$).

B) Usable Hosts per Subnet

With a /22 mask, we have 10 bits remaining for hosts ($32 - 22 = 10$).

Total Addresses: $2^{10} = 1,024$

Usable Host IPs: $1,024 - 2 = 1,022$ usable IP addresses per subnet.

C) Details for the 2nd Subnet

To find the range, we use the "magic number" in the third octet. Since our mask ends in .252, the increment is $256 - 252 = 4$. The subnets count up by 4 in the third octet.

1st Subnet ID 172.16.0.0

2nd Subnet ID 172.16.4.0

First Usable IP 172.16.4.1

Last Usable IP 172.16.7.254

Broadcast Address 172.16.7.255

Scenario 7: The Corporate Headquarters

The Situation: A multinational corporation uses the block **172.25.0.0/16** for its main headquarters. Rather than focusing on the number of subnets, the lead engineer dictates that every subnet must be sized to support exactly 500 usable host addresses to account for future employee growth per department. **Student Tasks:**

- A) How many host bits are required to support at least 500 usable hosts?
- B) What is the new subnet mask? (Provide CIDR and Decimal).
- C) What is the "Block Size" (magic number) in the 3rd octet for these subnets?

A) Host Bits Required

To find the number of bits (n) needed to support at least 500 usable hosts, we use the formula $2^n - 2 \geq 500$. If we use 8 bits: $2^8 - 2 = 254$ (Too small)

If we use 9 bits: $2^9 - 2 = 510$ usable hosts

B) The New Subnet Mask

Since an IPv4 address has 32 bits total, and we have reserved 9 bits for the hosts, the remaining bits belong to the network ($32 - 9 = 23$).

CIDR: /23

Dotted Decimal: 255.255.254.0

Why .254.0? The 23rd bit sits in the third octet. In binary, a /23 mask looks like 11111111.11111111.11111110.00000000. The sum of the bits in that third octet ($\$128 + 64 + 32 + 16 + 8 + 4 + 2$) equals 254.

C) The "Block Size" (Magic Number)

The "Magic Number" tells you the mathematical increment between your subnets. Finding this by subtracting the interesting octet (the one that isn't 0 or 255) from 256

$$256 - 254 = 2$$

Block Size: 2 in the 3rd octet.

Dept 1: 172.25.0.0

Dept 2: 172.25.2.0

Dept 3: 172.25.4.0

Dept 4: 172.25.6.0

Scenario 8: The Smart City Project

The Situation: A city government is rolling out a "Smart City" initiative using the network **128.10.0.0/16**. They need to create 100 subnets to cover different city grids, traffic light systems, and public Wi-Fi zones. **Student Tasks:**

- A) What subnet mask is required to create at least 100 subnets? (Provide CIDR and Decimal).
- B) How many usable IP addresses will be available in each city grid subnet?
- C) Provide the Network ID for the **3rd subnet**.

A) The Required Subnet Mask

To find the number of borrowed bits, we look for the smallest power of 2 that is greater than or equal to 100.

$$2^6 = 64 \text{ (Too small)}$$

$$2^7 = 128 \text{ (Perfect)}$$

By borrowing 7 bits from the original /16 network, we calculate the new mask:

CIDR: /23 (16 + 7 = 23)

Dotted Decimal: 255.255.254.0

B) Usable IP Addresses per Subnet

With a /23 mask, we have 9 bits remaining for hosts ($32 - 23 = 9$). We use the standard $2^n - 2$ formula:

Total Addresses: $2^9 = 512$

Usable Host IPs: $512 - 2 = 510$ usable IP addresses per city grid.

C) Network ID for the 3rd Subnet

To find the Network IDs, we use the "magic number" (increment) in the third octet. Since our mask value is 254 in that octet, the increment is $256 - 254 = 2$.

1st Subnet: 128.10.0.0

2nd Subnet: 128.10.2.0

3rd Subnet: 128.10.4.0

Network ID for the 3rd Subnet: 128.10.4.0

Scenario 9: The University Wi-Fi Zones

The Situation: A university is redesigning its wireless network using the block **172.31.0.0/16**. They have massive open-air study zones. The network architect requires that each subnet be large enough to support 2,000 wireless devices simultaneously. **Student**

Tasks:

- A) What subnet mask is required to support at least 2,000 usable hosts per subnet? (Provide CIDR and Decimal).
- B) How many total subnets does this configuration allow the university to create from their /16 block?
- C) Provide the Usable IP Range for the **1st subnet**.

A) The Required Subnet Mask

To support at least 2,000 usable hosts, we need to find the smallest power of 2 ($2^n - 2$) that covers that number.

$2^{10} - 2 = 1,022$ (Too small)

$2^{11} - 2 = 2,046$ (Perfect)

Since we need 11 bits for the host portion, we subtract that from the total 32 bits to find our network prefix ($32 - 11 = 21$).

CIDR: /21

Dotted Decimal: 255.255.248.0

B) Total Subnets Available

To find the number of subnets, we look at how many bits were borrowed from the original /16 network. We moved from /16 to /21, which means we borrowed 5 bits.

Formula: $2^5 = 32$

Total Subnets: This configuration allows for 32 total subnets.

C) Usable IP Range for the 1st Subnet

The "magic number" (increment) for a /21 mask is found in the third octet. Since $256 - 248 = 8$, the subnets will jump by 8 in the third octet.

For the 1st Subnet:

Network ID: 172.31.0.0

First Usable IP: 172.31.0.1

Last Usable IP: 172.31.7.254

Broadcast Address: 172.31.7.255

Scenario 10: The Manufacturing Plant

The Situation: A major automotive manufacturing plant uses the network **150.20.0.0/16**. The network must be divided into exactly 16 equal-sized subnets for different assembly lines and robotic control sectors. **Student Tasks:**

- A) What subnet mask will divide the network into exactly 16 subnets? (Provide CIDR and Decimal).
- B) How many usable hosts are available per assembly line?
- C) Provide the Network ID, Usable IP Range, and Broadcast Address for the **5th subnet**.

A) The Required Subnet Mask

To create exactly 16 subnets, we need to find the power of 2 that equals 16 ($2^n = 16$).

$$2^4 = 16.$$

We must borrow 4 bits from the original /16 host portion.

CIDR: /20 ($16 + 4 = 20$)

Dotted Decimal: 255.255.240.0

B) Usable Hosts per Assembly Line

With a /20 mask, we have 12 bits remaining for the host portion ($32 - 20 = 12$). We calculate the usable addresses using the $2^n - 2$ formula:

Total Addresses: $2^{12} = 4,096$

Usable Host IPs: $4,096 - 2 = 4,094$ usable IP addresses per subnet.

C) Details for the 5th Subnet

To find the range, we identify the "magic number" (the increment). Since the third octet of the mask is 240, the increment is $256 - 240 = 16$.

1st Subnet: 150.20.0.0

2nd Subnet: 150.20.16.0

3rd Subnet: 150.20.32.0

4th Subnet: 150.20.48.0

5th Subnet: 150.20.64.0

Breakdown for the 5th Subnet:

Network ID: 150.20.64.0

Usable IP Range: 150.20.64.1 to 150.20.79.254

Broadcast Address: 150.20.79.255

